Computation of Dynamic Models with Applications

GENERAL INFORMATION

Professor Minsu Chang

Class Time and Location: Wednesdays 9:30AM - 12:00PM at Car Barn 250.

- There will be no class on November 23rd.
- The last class will be on November 30th.
- Classes will take place *in-person* at their normal times and in their normal location unless I notify that I am implementing the instructional continuity plan (remote).
- Masks are required during in-person classes.

Office Hours and Location: Wednesdays 4:00PM - 5:00PM at ICC 553. (Sign-up required at `https://docs.google.com/spreadsheets/d/1Ar7YqFByHple2YI9mYz-3lShfZv3rFh2lDWC8G5Bn4o/edit#gid=0`.)

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COURSE DESCRIPTION

Course Website: Canvas page at `https://georgetown.instructure.com/courses/153142`.

Course Description: This course is an introduction to computing dynamic models in economics. We will study tools to solve and estimate linear/nonlinear models that are useful to conduct substantive quantitative research. Although we will mostly cover applications in macroeconomics, the methods taught throughout the course will be applicable to various dynamic models in economics.

Prerequisites: Graduate level macroeconomics and econometrics.

COURSE REQUIREMENTS

Problem Sets [40%]: There will be four problem sets, assigned during the semester. Students can discuss the problem sets, but each student has to write his/her own solution and codes. The solution together with the codes must be submitted via Canvas by 6:00 pm on the specified due dates (TBD). These due dates are binding and late submissions will not be accepted.

Class Discussions/Participation [30%]: After each problem set is submitted, it will be discussed in class by students. A student will be asked to present his/her answer to a problem set question. This will take place using zoom in which a student could share his/her screen. If there is a volunteer to present his/her results, extra credit will be given to the person. A typical presentation will include the main result and the summary of computational approach taken (highlighting key parts of one's code, not going through it line by line).

Paper Presentation [30%] Students will be assigned research papers on recent advances in computational methods. Each group of students[1] will have 20-30 minutes to present including questions from the audience. Then each presentation will be followed by a 5-minute discussion session. Students should send their slides to me by 6:00 pm the day before their presentation. A presentation should include the contribution of a paper and explain the gist of the methodology so that everyone in class has something learned from each presentation. Your own presentation will account for 25%. The remaining 5% is determined based on your participation and discussion regarding other students' presentations.

- October 26th   List of papers uploaded (paper suggestion by October 25th)
- November 2nd   Papers assigned
- November 30th   Presentations (in the format of a mini-conference)

[1] Depending on enrollment, it could be a team of two or three students.

TEXTBOOK AND SOFTWARE

Course Text: There is no official textbook for this course. My lecture slides, which will be posted online at the start of each week, are the final authority on course material.

Required Software: This course requires students to program in at least one programming language. MATLAB is recommended because it is commonly used in economics and some questions in problem sets will require software provided in MATLAB. I will talk more about this on the first day of class.

Recommended Texts: The following books and articles are suggested for your reference.

- Judd, Kenneth (1998): *Numerical Methods in Economics*, MIT Press.
- Marimon, Ramon and Andrew Scott (1999): *Computational Methods for the Study of Dynamic Economies*, Oxford University Press.
- Herbst, Edward and Frank Schorfheide (2015): *Bayesian Estimation of DSGE Models*, Princeton University Press.
- Fernandez-Villaverde, Jesus, Juan Rubio-Ramirez, and Frank Schorfheide (2016): *Solution and Estimation Methods for DSGE Models*, In: H. Uhlig and J. Taylor (eds.): Handbook of Macroeconomics, Vol 2, Elsevier.
- Krueger, Dirk, Kurt Mitman, and Fabrizio Perri (2016): *Macroeconomics and Household Heterogeneity*, In: H. Uhlig and J. Taylor (eds.): Handbook of Macroeconomics, Vol 2, Elsevier.

I will also refer to a list of published articles and working papers in our lectures.

COURSE OUTLINE

- **Numerical optimization and a brief intro to Bayesian inference**:
  optimization methods with/without derivatives, mixed methods, simulation methods, Bayes theorem, Bayesian linear regression

- **Linear(ized) models and estimation**:
  linearization with a basic RBC model, solving a linearized system, estimation with Kalman filter

- **Value Function Iterations**:
  VFI with finite/infinite horizon, numerical implementation, discretization of stochastic processes, convergence assessment, endogenous grid method

- **Nonlinear models**:
  examples of nonlinear models, intro to functional equation

- **Perturbation methods:**
  perturbation with a RBC model, higher moment approximation, non-local accuracy test, perturbation on value function

- **Projection methods:**
  Chebyshev polynomials, multidimensional problem

- **Estimation with particle filter:**
  nonlinear state-space model, bootstrap particle filter, generic particle filter, application with a small-scale DSGE model

- **Heterogeneous-agent models:**
  models without aggregate uncertainty, unexpected aggregate shock and transition dynamics, models with aggregate uncertainty

- **Recent advances in solution and estimation methods:**
  presentations by students